Field Trial Report: Rice Variety VR-7 'Varuna'

Chhattisgarh Agri Research Station (fictional) | Kharif Season 2025 | Lead: Dr. Anjali Verma

1. Objective

The trial compared the new drought-tolerant rice variety VR-7 (Varuna) against the common local variety Swarna across three districts, to measure yield, water use and days to maturity.

2. Method

Each district had 12 plots of 0.5 hectare each: 6 with VR-7 and 6 with Swarna. Sowing was done between 18 and 25 June 2025. Irrigation was cut by 30% for all plots to simulate a dry season.

3. Results

District	VR-7 yield (t/ha)	Swarna yield (t/ha)	VR-7 days to maturity
Durg	5.2	4.1	118
Bilaspur	4.9	3.8	121
Bastar	4.4	3.2	125
Average	4.83	3.70	121

On average VR-7 gave 30.5% higher yield than Swarna and used about 22% less water per kilogram of grain. Bastar had the lowest yield because of a 19-day dry spell in August.

4. Pest and Disease

VR-7 showed moderate resistance to blast disease. Brown planthopper attacks were similar in both varieties, so standard pest control is still needed.

5. Recommendation

VR-7 is recommended for rain-fed and low-irrigation areas. Seed will be available to farmers from the 2027 Kharif season at a planned price of INR 62 per kg. A second trial on salt-affected soil is planned for 2026.